

Sampling

Signals and Linear Systems

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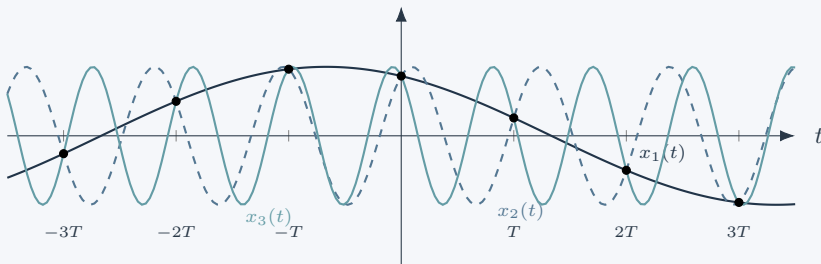
- **Introduction:** Representation of a Continuous-Time Signal by Its Samples
- **Impulse-Train Sampling:** Mathematical Model and Derivation
- **The Sampling Theorem:** Nyquist Rate and Aliasing
- **Signal Reconstruction:**
 - Ideal Low-Pass Filtering
 - Zero-Order Hold (ZOH)
 - Linear Interpolation
- **Revisiting the DFT:** Practical considerations and periodic signals

Introduction and Terminology

- In our review of the DFT, we established that evaluating a continuous-time signal $x(t)$ at uniform intervals T yields a discrete sequence $x[n] = x(nT)$.
- **Question:** Under what conditions can a continuous-time signal be uniquely determined by and exactly recovered from its discrete samples?
- This exact recovery is possible if the signal is **band-limited** (its Fourier Transform is zero outside a finite band of frequencies) and if the samples are taken sufficiently close together.
- This fundamental concept is known as the **Nyquist-Shannon Sampling Theorem**.

Ambiguity in Representation by Samples

- Generally, infinitely many continuous-time signals can yield the exactly same discrete sequence when sampled at interval T .

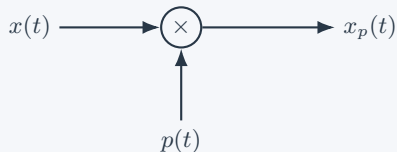


- To analyze sampling mathematically, it is convenient to represent the sampling process as multiplying the continuous signal $x(t)$ by a periodic impulse train $p(t)$.

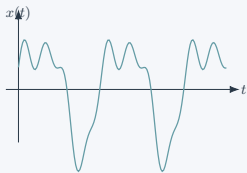
Impulse-Train Modulator

$$p(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT)$$

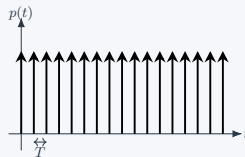
$$x_p(t) = x(t)p(t) = \sum_{n=-\infty}^{+\infty} x(nT)\delta(t - nT)$$



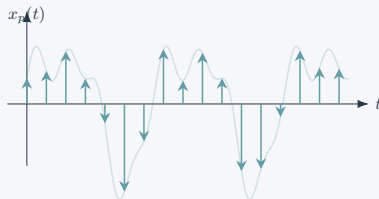
Visualizing Impulse-Train Sampling



Continuous signal $x(t)$.



Impulse train $p(t)$.



Sampled signal $x_p(t)$.

Derivation of $X_p(j\omega)$

- To see the effect of sampling, we need the Fourier Transform of $x_p(t) = x(t)p(t)$.
- Let's briefly derive the **duality principle**. From the inverse Fourier Transform:

$$x(t) = \frac{1}{2\pi} \int X(j\omega) e^{j\omega t} d\omega \implies 2\pi x(-t) = \int X(j\omega) e^{-j\omega t} d\omega$$

- Swapping t and ω :

$$2\pi x(-\omega) = \int X(jt) e^{-j\omega t} dt = \mathcal{F}\{X(jt)\}$$

- Hence, the duality principle is as follows:

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega) \implies X(jt) \xleftrightarrow{\mathcal{F}} 2\pi x(-\omega)$$

Multiplication is Convolution in the Frequency Domain

- We know from the Convolution Property that,

$$x_1(t) * x_2(t) \xleftrightarrow{\mathcal{F}} X_1(j\omega)X_2(j\omega)$$

- By duality,

$$X_1(jt) * X_2(jt) \xleftrightarrow{\mathcal{F}} 2\pi x_1(-\omega)x_2(-\omega)$$

$$X_1(jt) \xleftrightarrow{\mathcal{F}} 2\pi x_1(-\omega)$$

$$X_2(jt) \xleftrightarrow{\mathcal{F}} 2\pi x_2(-\omega)$$

- Substituting $X_1(jt)$ with $u(t)$ and $X_2(jt)$ with $v(t)$:

$$u(t) \xleftrightarrow{\mathcal{F}} 2\pi x_1(-\omega) = U(j\omega)$$

$$v(t) \xleftrightarrow{\mathcal{F}} 2\pi x_2(-\omega) = V(j\omega)$$

- Substituting we get:

$$u(t)v(t) \xleftrightarrow{\mathcal{F}} \frac{1}{2\pi} U(j\omega) * V(j\omega)$$

- Hence, we can write the Fourier Transform of the sampled signal as:

$$X_p(j\omega) = \frac{1}{2\pi} [X(j\omega) * P(j\omega)]$$

- Directly taking the Fourier Transform results in a summation that is hard to work with. Instead, we can use the Fourier Series.
- $p(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT)$ is clearly periodic with period T and thus using Fourier Series:

$$p(t) = \sum_{k=-\infty}^{+\infty} c_k e^{jk\omega_s t}, \quad \text{where } \omega_s = \frac{2\pi}{T}.$$

- Using the **sifting property** $\int f(t)\delta(t)dt = f(0)$ on period $[-T/2, T/2]$:

$$c_k = \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_s t} dt = \frac{1}{T}$$

- Hence, we can write:

$$p(t) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} e^{jk\omega_s t}$$

- We know the FT of an impulse is 1. By duality, the FT of a constant is an impulse.

$$\delta(t) \xleftrightarrow{\mathcal{F}} 1$$
$$1 \xleftrightarrow{\mathcal{F}} 2\pi\delta(\omega)$$

- Using the phase-shifting property:

$$e^{j\omega_0 t} x(t) \xleftrightarrow{\mathcal{F}} \int_{-\infty}^{+\infty} x(t) e^{-j(\omega - \omega_0)t} dt = X(j(\omega - \omega_0))$$

- Thus, we have:

$$e^{jk\omega_s t} \xleftrightarrow{\mathcal{F}} 2\pi\delta(\omega - k\omega_s)$$

- Taking the Continuous-Time Fourier Transform of $p(t)$ term-by-term yields another impulse train:

$$P(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} \mathcal{F}\{e^{jk\omega_s t}\} = \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s)$$

Spectrum of the Sampled Signal

- Now we evaluate the convolution $X_p(j\omega) = \frac{1}{2\pi} X(j\omega) * P(j\omega)$:

$$\begin{aligned} X_p(j\omega) &= \frac{1}{2\pi} X(j\omega) * \left[\frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s) \right] \\ &= \frac{1}{T} \sum_{k=-\infty}^{+\infty} X(j\omega) * \delta(\omega - k\omega_s) \end{aligned}$$

- For convolution with a shifted impulse, we use the integral definition and sifting property:

$$X(j\omega) * \delta(\omega - \omega_0) = \int_{-\infty}^{+\infty} X(j\theta) \delta(\omega - \theta - \omega_0) d\theta = X(j(\omega - \omega_0))$$

- Thus, we obtain the fundamental equation of sampling:

Spectrum of Sampled Signal

$$X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} X(j(\omega - k\omega_s))$$

- $X_p(j\omega)$ consists of periodically repeated copies of $X(j\omega)$, scaled by $1/T$.

The Sampling Theorem

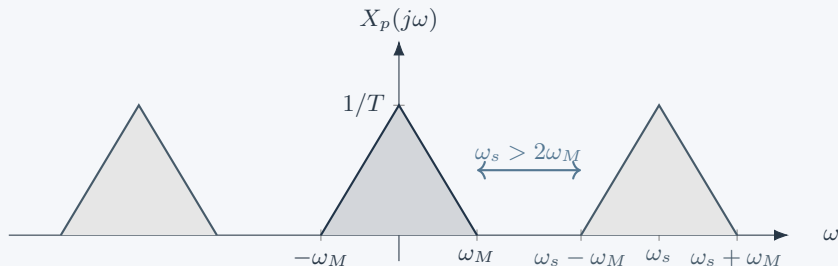
- Suppose $x(t)$ is a **band-limited** signal, meaning $X(j\omega) = 0$ for $|\omega| > \omega_M$.
- The copies of $X(j\omega)$ in $X_p(j\omega)$ are centered at integer multiples of ω_s .
- To avoid overlap between adjacent copies, the separation ω_s must be strictly greater than twice the maximum frequency ω_M .

Nyquist-Shannon Sampling Theorem

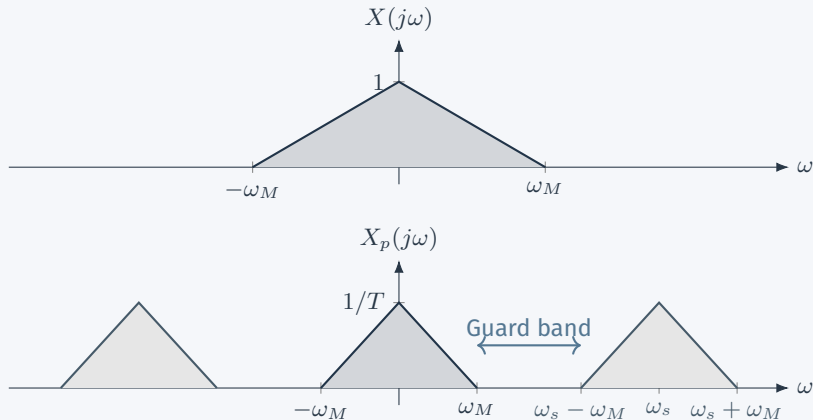
Let $x(t)$ be a band-limited signal with $X(j\omega) = 0$ for $|\omega| > \omega_M$. Then $x(t)$ is uniquely determined by its samples $x(nT)$, $n = 0, \pm 1, \pm 2, \dots$, if

$$\omega_s > 2\omega_M$$

where $\omega_s = 2\pi/T$. The minimum sampling frequency $2\omega_M$ is called the **Nyquist rate**.

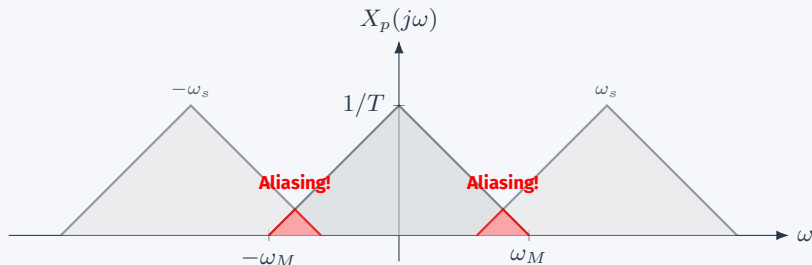


Visualizing the Spectrum: Oversampling



- When $\omega_s > 2\omega_M$, exact copies of the original spectrum are perfectly separated.
- The original signal can be recovered by isolating the baseband copy using a low-pass filter!

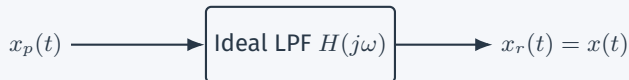
Visualizing the Spectrum: Undersampling (Aliasing)



- When $\omega_s < 2\omega_M$, the shifted copies overlap with the baseband spectrum.
- High frequencies fold over into the low-frequency band, distorting the signal. This phenomenon is called **aliasing**.
- The original signal $x(t)$ cannot be recovered by low-pass filtering.

Signal Reconstruction

- By the sampling theorem, if $\omega_s > 2\omega_M$, the **baseband spectrum** of $x_p(t)$ (the copy of the spectrum $X(j\omega)$ at the origin) is an exact replica of $X(j\omega)$ scaled by $1/T$.
- We can recover $x(t)$ by applying an **ideal low-pass filter** to $x_p(t)$.

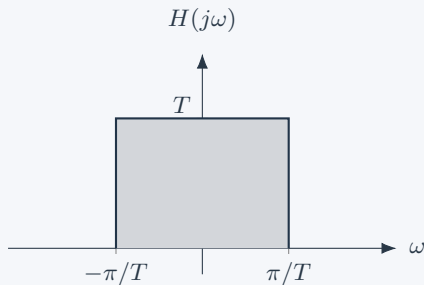


- The frequency response of the ideal LPF must be:

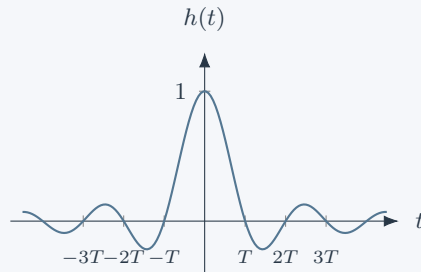
$$H(j\omega) = \begin{cases} T & |\omega| < \omega_c \\ 0 & |\omega| > \omega_c \end{cases} \quad \text{where} \quad \omega_M < \omega_c < \omega_s - \omega_M$$

- Commonly, we choose the cutoff frequency exactly in the middle: $\omega_c = \omega_s/2$, because it follows the assumption that $\omega_M < \omega_s/2$.

The Ideal Low-Pass Filter



Frequency Response $H(j\omega)$



Impulse Response $h(t)$

- The frequency response $H(j\omega)$ is a rectangular pulse with cutoff $\omega_c = \omega_s/2 = 2\pi f_s/2 = \pi/T$.
- The time-domain impulse response $h(t) = \text{sinc}(t/T)$ extends infinitely in both directions, making the ideal LPF **unrealizable**.

To find the time-domain impulse response $h(t)$ of an ideal low-pass filter with cutoff $\omega_c = \pi/T$ and constant amplitude T , we compute the Inverse Fourier Transform:

$$\begin{aligned} h(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} H(j\omega) e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-\pi/T}^{\pi/T} T e^{j\omega t} d\omega \\ &= \frac{T}{2\pi} \left[\frac{e^{j\omega t}}{jt} \right]_{-\pi/T}^{\pi/T} = \frac{T}{\pi t} \left(\frac{e^{j(\pi/T)t} - e^{-j(\pi/T)t}}{2j} \right) \\ &= \frac{T}{\pi t} \sin\left(\frac{\pi t}{T}\right) = \text{sinc}\left(\frac{t}{T}\right) \quad \left[\text{Recall } \text{sinc}(x) = \frac{\sin(\pi x)}{\pi x} \right] \end{aligned}$$

Result: The time-domain filter corresponding to a perfect frequency cutoff is exactly the infinitely wide sinc function!

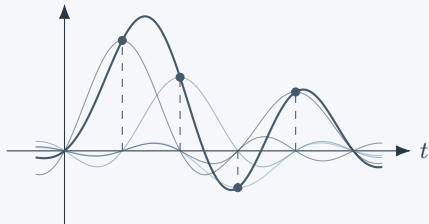
- The impulse response $h(t)$ of the ideal LPF (cutoff $\omega_c = \pi/T$) is:

$$h(t) = \text{sinc}\left(\frac{t}{T}\right) = \frac{\sin(\pi t/T)}{\pi t/T}$$

- The reconstructed signal $x_r(t) = x_p(t) * h(t)$ becomes:

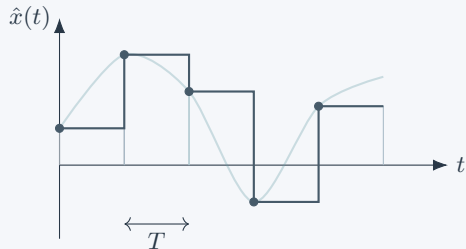
$$\begin{aligned} x_r(t) &= \left(\sum_{n=-\infty}^{+\infty} x(nT) \delta(t - nT) \right) * h(t) \\ &= \sum_{n=-\infty}^{+\infty} x(nT) \text{sinc}\left(\frac{t - nT}{T}\right) \end{aligned}$$

- Superposition of shifted sinc functions.
- However**, $h(t)$ is non-causal and decays very slowly. It is **unrealizable** in practice. We need approximations.



The simplest practical reconstruction method is the **Zero-Order Hold (ZOH)**.

- **Concept:** Simply hold the sample value constant until the next sample arrives.
- This forms a “staircase” approximation $x_0(t)$ of the continuous signal.
- Widely used in Digital-to-Analog Converters (DACs).



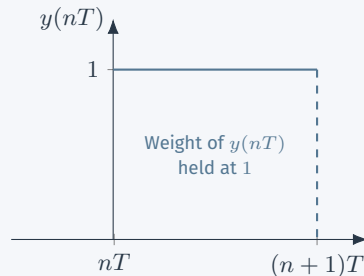
- We can view this mathematically as filtering the impulse-sampled signal $x_p(t)$ with a rectangular pulse $h_0(t)$.

Why is $h_0(t)$ a Rectangle?

- In a reconstructed output signal, an original sample $y(nT)$ contributes strictly via a shifted, scaled version of the base filter: $y(nT) \cdot h(t - nT)$.
- The shape of $h(t)$ dictates the **weighting** of the sample $y(nT)$ over continuous time.

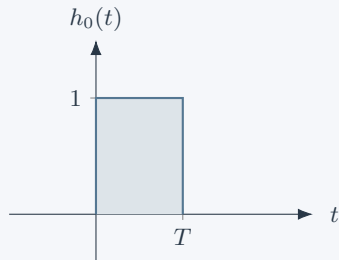
For Zero-Order Hold:

- We desire the sample $y(nT)$ to simply hold its full weight of 1 constant precisely until the next sample at $(n + 1)T$ arrives.
- Thus, the isolated weight of $y(nT)$ must jump to 1 at $t = nT$ and abruptly drop back to 0 at $t = (n + 1)T$.
- Therefore, the underlying impulse response $h_0(t)$ **must** inherently be a rectangular block formally spanning from 0 to T !

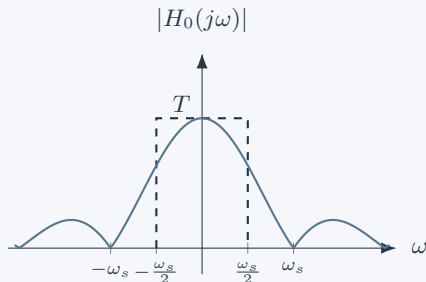


The ZOH impulse response $h_0(t)$ is simply a rectangular pulse equal to 1 over $[0, T]$. We construct $H_0(j\omega)$ strictly via continuous Fourier integration:

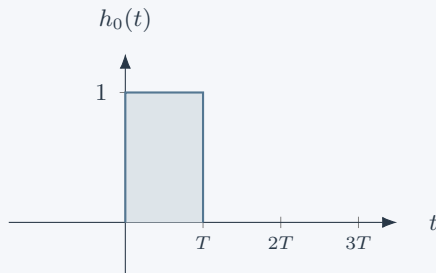
$$\begin{aligned} H_0(j\omega) &= \int_{-\infty}^{\infty} h_0(t) e^{-j\omega t} dt \\ &= \int_0^T (1) e^{-j\omega t} dt = \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^T \\ &= \frac{1 - e^{-j\omega T}}{j\omega} = \frac{e^{-j\omega T/2} (e^{j\omega T/2} - e^{-j\omega T/2})}{j\omega} \\ &= e^{-j\omega T/2} \left[\frac{2}{\omega} \sin \left(\frac{\omega T}{2} \right) \right] \\ &= T e^{-j\omega T/2} \text{sinc} \left(\frac{\omega T}{2\pi} \right) \end{aligned}$$



Frequency Domain view of ZOH



Frequency Response $|H_0(j\omega)|$



Impulse Response $h_0(t)$

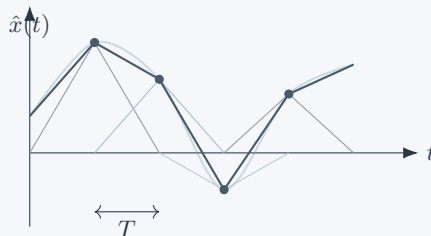
- $h_0(t)$ is a rectangle of width T . Its frequency response includes a sinc function:
$$H_0(j\omega) = T e^{-j\omega T/2} \text{sinc}\left(\frac{\omega T}{2\pi}\right).$$

We notice that,

- In the passband $|\omega| \leq \omega_s/2$, the gain is not constant, attenuating high frequencies slightly.
- Outside the passband, the gain is non-zero, allowing high-frequency aliasing copies to leak through.

Linear Interpolation (First-Order Hold)

- **Linear Interpolation** connects adjacent samples with straight lines.
- This produces a continuous, much smoother approximation $x_1(t)$ than the ZOH staircase.
- Achieved by filtering the impulse-sampled signal $x_p(t)$ with a triangular pulse $h_1(t)$.



- The impulse response $h_1(t)$ is a triangle spanning $[-T, T]$ with peak height 1 at $t = 0$.

Why is $h_1(t)$ a Triangle?

- In a reconstructed signal, a given sample $y(nT)$ contributes via $y(nT) \cdot h_1(t - nT)$.
- The shape of $h_1(t)$ thus dictates the **weighting** of $y(nT)$ over time.

For Linear Interpolation:

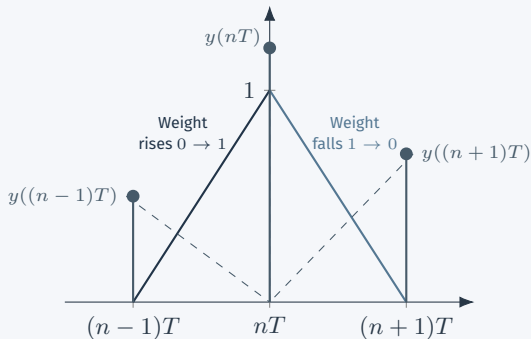
Given a weight $\alpha \in [0, 1]$, we should have,

$$y(t) = (1 - \alpha)y((n - 1)T) + \alpha y(nT) \quad \text{for } t \in [(n - 1)T, nT]$$

$$y(t) = (1 - \alpha)y(nT) + \alpha y((n + 1)T) \quad \text{for } t \in [nT, (n + 1)T]$$

Therefore,

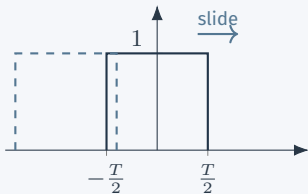
- The weight of $y(nT)$ should linearly rise strictly from 0 to 1 between $(n - 1)T$ and nT .
- It should linearly fall from 1 back to 0 between nT and $(n + 1)T$.
- At all other times, its weight must precisely be 0 to isolate its exact local influence.
- Therefore, $h_1(t)$ **must** inherently be a symmetric triangle!



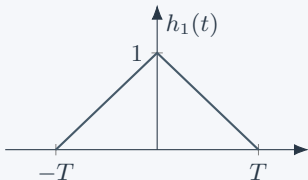
A linear interpolation triangle filter $h_1(t)$ can be formed by convolving two shifted zero-order hold rectangles!

$$h_1(t) = \frac{1}{T} \left(h_0(t + T/2) * h_0(t + T/2) \right)$$

Rectangles overlapping

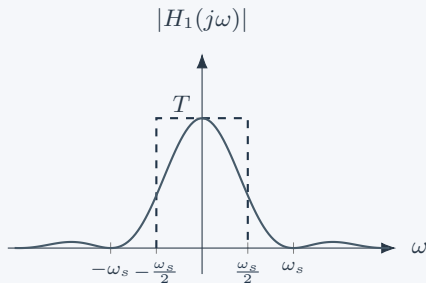


- $h_0(t + T/2)$ is just the standard ZOH block, centered exactly on time $t = 0$.
- Area of overlap grows linearly to a peak, and drops cleanly linearly!
- The peak area is $T \times 1 = T$, hence the factor $\frac{1}{T}$, as the triangle peak should be 1.

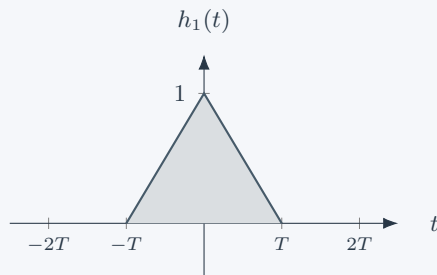


Frequency Derivation: Use $\mathcal{F}\{x * x\} = \mathcal{F}\{x\}^2$.

$$\begin{aligned} H_1(j\omega) &= \frac{1}{T} \left[\mathcal{F}\{h_0(t + T/2)\} \right]^2 \\ &= \frac{1}{T} \left[T \operatorname{sinc}\left(\frac{\omega T}{2\pi}\right) \right]^2 \\ &= T \operatorname{sinc}^2\left(\frac{\omega T}{2\pi}\right) \end{aligned}$$



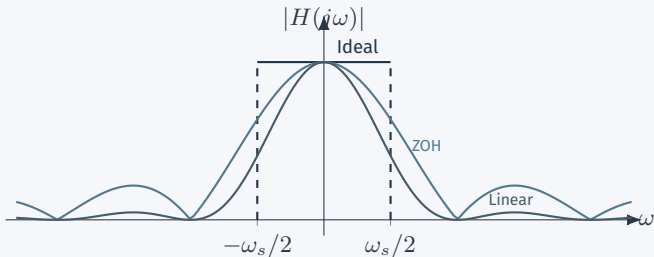
Frequency Response $|H_1(j\omega)|$



Impulse Response $h_1(t)$

- $h_1(t)$ is a triangular pulse spanning $[-T, T]$. Its frequency response is $H_1(j\omega) = T \text{sinc}^2\left(\frac{\omega T}{2\pi}\right)$.
- Because of the convolution property, $H_1(j\omega)$ is proportional to the square of the ZOH response, without the phase delay.

- Let's compare the frequency domain magnitudes against the ideal filter (assuming $T = 1$):



- Linear interpolation has much faster high-frequency rolloff (less leakage) than ZOH, making it a "better" low-pass filter approximation.

Connection to the DFT

The Uncertainty Principle

Signals **cannot** be both time-limited and strictly band-limited!

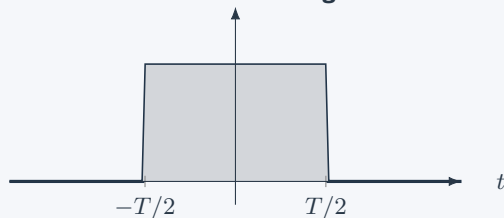
- Finite-duration captures mean $X(j\omega)$ inherently extends to infinite frequencies.^a

^aInteresting watch: 3Blue1Brown - Uncertainty Principle

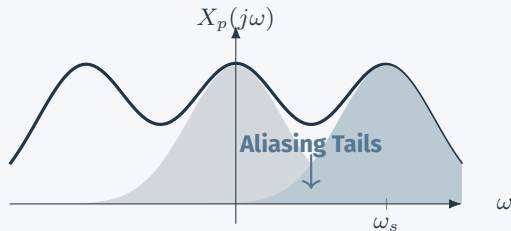
Unavoidable Aliasing

Sampling any real-world signal causes some aliasing! The infinite spectral tails will always overlap and leak into our baseband.

Time-Limited Signal



Infinite Frequency Spectrum & Aliasing



DTFT

The DTFT is exactly the Fourier transform of the impulse train $x_p(t)$ as we saw earlier!

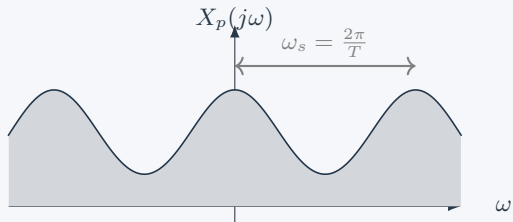
What does the DFT do?

The N -point DFT simply grabs N evenly spaced samples of this spectrum from 0 to ω_s :

$$X[k] = X_p\left(j\frac{k\omega_s}{N}\right)$$

The rightmost bins naturally wrap back negative frequencies!

Transform of periodically sampled $x(t)$



Discrete Fourier Transform

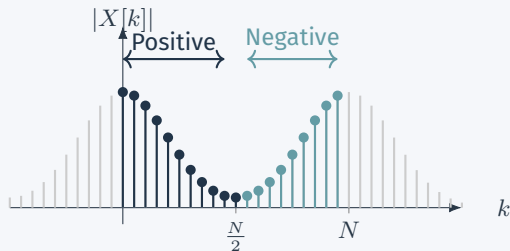


How to Read the DFT Bins

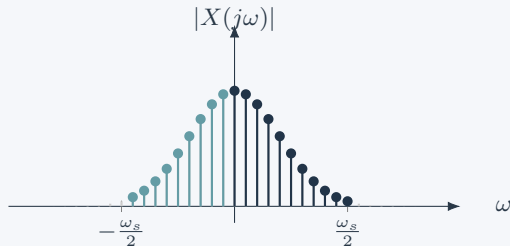
We compute from $0 \rightarrow \omega_s$, but physical signals are centered around 0. Because the spectrum repeats (periodicity), we just "fold" the second half back!

- **First Half: Positive** frequencies ($0 \leq k < N/2$).
- **Second Half: Negative** frequencies ($N/2 < k < N$).

DFT Bin Index Mapping



Equivalent Baseband Spectrum



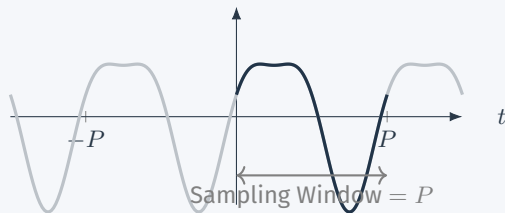
Periodic Signals $\equiv$ Deltas

If our signal repeats perfectly forever with period P , it can be represented entirely by a Fourier Series!

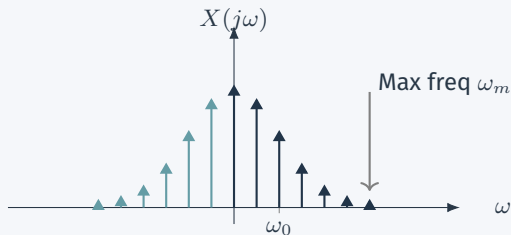
- Instead of a continuous smear, $X(j\omega)$ just becomes discrete **delta spikes** exactly at harmonics (ω_0):

$$X(j\omega) = 2\pi \sum_{m=-\infty}^{\infty} a_m \delta(\omega - m\omega_0)$$

Time Domain: Periodic outside window



Frequency Domain: Sum of Deltas $X(j\omega)$



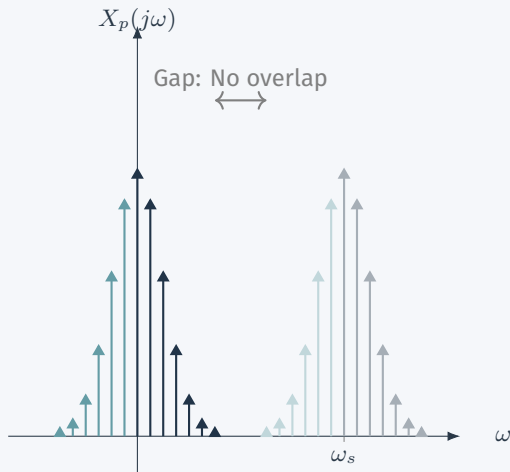
Preserving the Spikes (Nyquist)

Remember, sampling creates infinite shifted copies of our delta spectrum!

$$X_p(j\omega) = \frac{1}{T} \sum_{r=-\infty}^{\infty} X(j(\omega - r\omega_s))$$

As long as we sample fast enough ($\omega_s > 2\omega_m$), we leave a clear gap. **Zero shifted copies overlap**, which implies exactly **zero aliasing**!

Shifted Copies $X_p(j\omega)$ without Aliasing



Perfect Isolation

By capturing exactly N samples per period, our DFT evaluation bins **land perfectly** on top of our isolated deltas!

- We perfectly extract the true a_m amplitudes without any distortion!

$$X[k] = X_p\left(j\frac{k\omega_s}{N}\right) \propto a_k$$

Positive a_m : Captured cleanly at the start bins.

Negative a_{-m} : Captured reliably at the folded tail!

Discrete Fourier Transform Evaluates $X_p(j\omega)$

